



Course Outline

1. Course Identity

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	GEB2405
Course title (English)	Understanding Data Science
Course title (Chinese)	认识数据科学
Units	3
Language of Instruction	English
Description (English)	We live in a world surrounded by data and technology. While China is massively investing in Artificial Intelligence, Data science is believed to be a key skill of the 21st century. Capturing, framing, and analyzing data is already a crucial challenge for companies and scientists. These skills are also critical for students since data science can be applied to discourse analysis, economic prediction, marketing, social sciences, and so forth. For those reasons, many universities in the US already offer "introduction to data science" as part of their general education program. As a data science GE class, this course provides the students with some basic proficiency in data analysis (organize, manage, examine, prepare, analyze, and visualize data using R studio). Students also discover how data science can help to understand social and natural phenomena in various domains by exploring real datasets (social science survey, titanic death toll, economic development indicators, health survey...etc).
Description (Chinese)	当今世界无不被数据和科技所包围。随着中国在人工智能方面大力投资，数据科学被认为是21世纪的一项关键技能。收集，和分析数据已经是公司和科学家的一项重要挑战。这些技能对学生也很重要，因为数据科学可以应用于话语分析，经济预测，市场营销，社会科学等等。基于这些原因，许多美国大学已开设本课程为"数据科学入门"通识课程。 作为通识教育数据科学课程，本课程为学生提供一些数据分析的基本知识（使用R studio组织、管理、检查、准备、分析和可视化数据）。通过探索真实的数据集（社会科学调查、泰坦尼克号死亡人数、经济发展指标、健康调查等数据），学生还将发现数据科学如何帮助理解各个领域的社会和自然现象。

2. Prerequisites / Co-requisites



A. Prerequisites

GFH1000/GFN1000

Since this course focus on programming aspects, there are no prerequisites. This course is designed for students having no previous experience with R.

B. Co-requisites

GFH1000/GFN1000

Students who have already learnt ERG2050 are not eligible for this class

Students in HSS are recommended to take HSS1009 "Mathematics (Humanities and Social Sciences)" before taking this class.

3. Learning Outcomes

By the end of the course, students should be able to:

3.1 Knowledge:

- Identify the correlation between two categorical variables through logistic regression and bivariate analysis
- Analyze various natural and social phenomena through statistical inference
- Know how to test the strength and the validity of logistic regression model

3.2 Skills

- Master core R programming skills
- Inspect, clean and prepare second-hand data for analysis
- Create effective visualizations of given data (to communicate or persuade).

3.3 Value

1. Understand how data analysis is crucial in many field of knowledge
2. Reflect critically on on the difference between correlation and causuation

4. Course syllabus

The course is divided into three sections.

Introduction to R and data preparation

The course starts with 12 teaching hours dedicated to "preparation of data for analysis".



This section cover contents and techniques such as "Write, save and launch simple commands", "finding and import second-hand data", "Prepare a paper based survey for analysis", "Inspect, clean and transform data", "Store data and results".

Create effective visualization.

The second section lasts for 8 teachings hours. Students learn how to produce cutting-edge figures such as bar charts, column charts, multiple bar charts and pie charts.

Using data to understand the society.

The last section lasts for 18 teaching hours. During the first 5 teaching hours, students learn how to program and analyse cross-tabulation. Then, students are introduced to the Simpson's paradox and learn how to apply, build, test and fit logistic regression model to specific research question (who did survive the titanic? How American students are choosing their college program....)

Class organization: Lecture, practicum, tutorial

Weekly planning:

1 h lecture: 60 students

1h online session (recorded lecture): 60 students

1h tutorial, 3 groups of 20 students

5. Assessment Scheme

Component/ method	% weight
Tests in class (week 5, week 8 and week 14)	60
Quizzes	15
Exercise sessions during the tutorial (week 6, 11)	25

6. Grade descriptor

General

Grade	Description
A	Outstanding performance on selecting, computing, combining the relevant R commands to deal with authentic data. Excellent capacity to produce cutting edge figures and to communicate results. Excellent capacity to verify statistical assumptions, and to identify statistical correlation. Outstanding technical and comprehensive understanding of Geometric data analysis and clustering basics. Accurate understanding of R outputs.
A-	Outstanding performance on selecting, computing, combining the relevant R commands to deal with authentic data. Most of the learning outcomes are brilliantly achieved while one of the areas should be ameliorated (capacity to produce cutting edge figures and to communicate results/ capacity to verify statistical assumptions, and to identify statistical / correlation/ technical and comprehensive understanding



	of Geometric data analysis and clustering basics/Precise and accurate understanding of R outputs)
B	Good performance on selecting, computing, combining the relevant R commands to deal with authentic data. Efficiency can be improved in some computing operations. Most of the learning outcomes are achieved while one of the areas should be ameliorated (capacity to produce cutting edge figures and to communicate results/ capacity to verify statistical assumptions, and to identify statistical / correlation/ technical and comprehensive understanding of Geometric data analysis and clustering basics/Precise and accurate understanding of R outputs)
C	Satisfactory performance on selecting, computing, combining the relevant R commands to deal with authentic data. Efficiency can be improved in some computing operations. Some of the learning outcomes are brilliantly achieved while several areas should be ameliorated (capacity to produce cutting edge figures and to communicate results/ capacity to verify statistical assumptions, and to identify statistical / correlation/ technical and comprehensive understanding of Geometric data analysis and clustering basics/Precise and accurate understanding of R outputs)
D	Borderline performance on selecting, computing, combining the relevant R commands to deal with authentic data. Efficiency in computing operations is average. Learning outcomes are poorly achieved while several areas should be ameliorated (capacity to produce cutting edge figures and to communicate results/ capacity to verify statistical assumptions, and to identify statistical / correlation/ technical and comprehensive understanding of Geometric data analysis and clustering basics/Precise and accurate understanding of R outputs)
F	Unsatisfactory performance on selecting, computing, combining the relevant R commands to deal with authentic data. Efficiency in computing operations is low and cannot be performed autonomously. Most of the learning outcomes are not achieved.

7. Feedback for evaluation

Online Course and Teaching Evaluation (OCTE)

8. Reading

Required Reading

1.R for data science, 1491910399, Hadley Wickha, O Reilly, 1st, 2017, United States, required

2.Quantitative Social Science with R, 9781526411501, Brian J. Fogarty, SAGE, 1st,



2018, United States, required

9. Course components

Activity	Hours/week
Lecture	1
Online training	1
Tutorial session	1

10. Indicative teaching plan

Week	Content/ topic/ activity
01	Lecture 1 Why learning R studio : a world of data Online 1 Getting prepared for your first step in the data world Tutorial 1 Discovering R environment, writing my first code: Writing information about people suffering from diabetes in R
02	Lecture 2 Working on second-hand data: Example of bias and challenges in opinion and consumption survey Online 2 How to get data: Exploration of foreign and Chinese database Tutorial 2 From the web to my computer: Importing, view and screening official data
03	Lecture 3 Why data manipulation is crucial and sensitive?: An example of the manipulation the China General Social Survey data for a research on lifestyle. Online 3 When data are messy: Prepare the titanic passengers dataset for further analysis Tutorial 3 From number to information: Editing the millennium cohort survey according to its codebook
04	Lecture 4 How and why social scientists like to work with paper based survey? Online 4 Merge the data collected by different interviewers: Example of an employment survey on graduate students Tutorial 4 Paper based questionnaire in practice: Exercise based on a survey on students coffee consumption.
05	Lecture 5 First Exam Online 5 Visualizing the relation between Body Mass Index and Socio-Economic Status Tutorial 5 Does English pupils have unhealthy habits? : Graphing the smoking and drug experience of the English youth
06	Lecture 6 How to produce meaningful data visualization? Good practices and some example of rookie mistakes Online 6 Graphing the relation between two variables: Examples from the an health survey and from motor car road test Tutorial 6 Visualize the relation between well-being, inequality and economic development
07	Lecture 7 Data visualization: When programming meets arts.



	Online 7 Using pie chart and subgraph to identify groups: Examples from the employment survey Tutorial 7 How students choose their class? Visualizing the choice of students depending on their major, gender and other characteristics
08	Lecture 8 Second exam Online 8 Understanding how gender influence earning and smoking habits through crosstabulation Tutorial 8 How does student choose their GE class? Explore the the choice of students depending on their major, gender and other characteristics through crosstabulation
09	Lecture 9 What does say a table and what we say about it: difference between correlation and causation Online 9 The association between gender, earning and diploma in the golf trainers survey: Looking behind Specific and general correlations. Tutorial 9 How parents are tutoring their child? Exploring the Millenium survey through crosstabulation
10	Lecture 10 From Simpson paradox to reading a model: How and why scientist use regression model? Online 10 Who can be a manager in the golf industry? An example of binary regression models building and fitting with R Tutorial 10 Who did survive the Titanic? An example of binary regression model building and fitting with R
11	Lecture 11 Who can be a manager in the golf industry? An example about how to make sure that a binary logistic model is not biased Online 11 When the assumptions are not met: Test and Practical consideration Tutorial 11 Making a good model: Exercise on heart disease
12	Lecture 12 How data science helps to follow and anticipate future health expenditure Online 12 The Choice of higher education program in the USA : How multinomial logistic regression can help to understand it? Tutorial 12 Who did vote for Trump in 2016?: Exercise on multinomial logistic regression using R
13	Lecture 13 How amazon and Jingdong are tempting you: Principle of machine learning and neural network Online 13 Online Quizz Tutorial 13 Explaining the taste and distate for metal music: An application of multinomial logistic regression to GBCS data
14	Lecture 14 Third exam Online 14 Make R studio fancy : what can be done with Rmarkdown Tutorial 14 Make R studio fancy : Interactive data presentation with R Shiny

11. Any other information

After a double benchmark, this course has been designed to make it conform to the GE goals and not redundant with other courses taught in CUHK Shenzhen.



It was designed after considering the syllabus of "introduction to data science" courses taught as part of general education at Berkley, Carnegie Mellon, Winona State University, and Montgomery College.

It was designed after considering several courses, including HSS1009, STA4030, STA3010, STA 3050, ERG2050, ECO2121, PSY5020, PSY5010. As discussed with the HSS department colleagues, the focus on programming and categorical data avoid overlap with those courses.

The students who have taken ERG2050 are not eligible for this class.

Attention is drawn to University policy and regulations on honesty in academic work, and to the disciplinary guidelines and procedures applicable to breaches of such policy and regulations. Details may be found at <https://registry.cuhk.edu.cn/en/page/30>

With each assignment, students will be required to submit a signed declaration that they are aware of these policies, regulations, guidelines and procedures. For group projects, all students of the same group should be asked to sign on the declaration.

12. Version date

Version number	1
As of (date)	2023/12/18